

Course: B. Tech. Branch: Electronics and Communication Engineering /
Electronics and Telecommunication Engineering Semester: IV
Subject Code & Name: Network Theory (BTETC401)

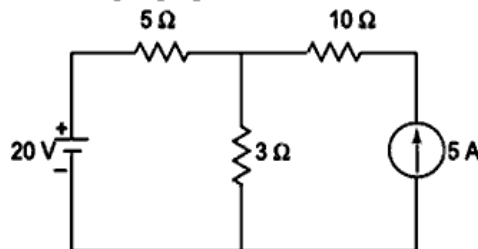
Max Marks: 60 Date:16-01-24 Duration: 3 Hr.

Instructions to the Students:

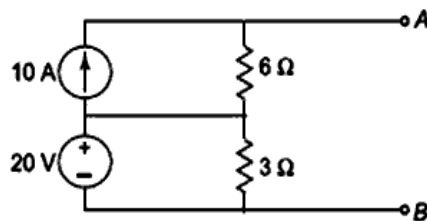
1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12

- | | | |
|---|-----|---|
| A) State Superposition theorem and find the current passing through the 3 Ω resistor in the circuit using superposition theorem | CO1 | 6 |
|---|-----|---|



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|--|-----|---|
| B) Replace the given network shown in Fig by a single current source in parallel with a resistance | CO1 | 6 |
|--|-----|---|

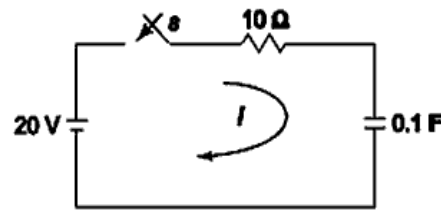


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|---|-----|---|
| C) Define the terms: TREE AND CO-TREE, Obtain the incidence matrix A from the following reduced incidence matrix A1 and draw its graph. | CO1 | 6 |
|---|-----|---|

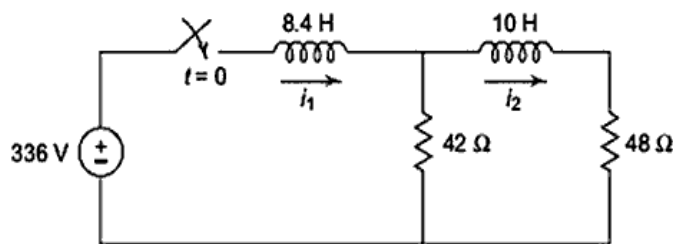
$$[A_1] = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 & 1 \end{bmatrix}$$

Q.2 Solve Any Two of the following.	12
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- A) A series RC circuit consists of resistor of $10\ \Omega$ and capacitor of 0.1 F as shown in Fig. A constant voltage of 20 V is applied to the circuit at $t = 0$. Obtain the current equation. Determine the voltages across the resistor and the capacitor. CO1, CO3 6

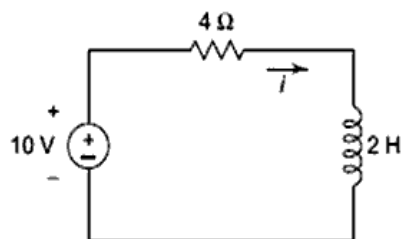


- B) Derive an expression for Dc response of an R-L circuit CO1, CO3 6
- C) Obtain the expression for i_1 and i_2 in the circuit shown in Fig. when dc voltage source is applied suddenly. Assume that the initial energy stored in the circuit is zero. CO1, CO3 6



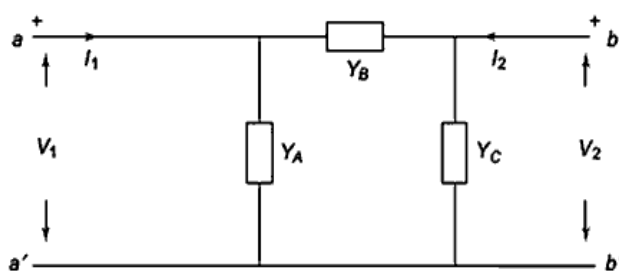
Q. 3 Solve Any Two of the following. 12

- A) Write Laplace transform of some standard Network Functions. CO1, CO3 6
- B) Explain Circuit elements in the s-domain CO1, CO3 6
- C) Determine the current i for $t \geq 0$ if initial current $i(0) = 1$ for the circuit shown in Fig. CO1 6



Q.4 Solve Any Two of the following. 12

- A) Find the Y parameters for the network shown in Fig. CO1, CO4 6



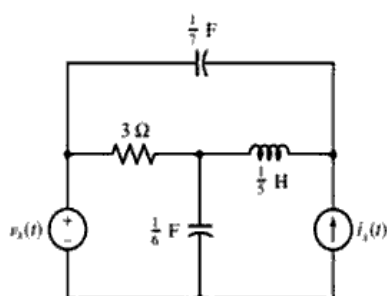
- B)** In the two-port network, compute h-parameters from the following data: **CO1, CO4** **6**
- (a) With the output port short-circuited: $V_1 = 25 \text{ V}$, $I_1 = 1 \text{ A}$, $I_2 = 2 \text{ A}$
- (b) With the input port open-circuited: $V_1 = 10 \text{ V}$, $V_2 = 50 \text{ V}$, $I_2 = 2 \text{ A}$
- C)** Derive Z-parameter in terms of Y-parameter, h-parameter and ABCD parameter **CO1, CO4** **6**

Q. 5 Solve Any Two of the following. **12**

- A)** Find the two Cauer realisations of driving point function given by **CO1, CO3** **6**

$$Z(S) = \frac{10S^4 + 12S^2 + 1}{2S^3 + 2S}$$

- B)** Obtain the normal-form equations for the circuit of Fig., a four-node circuit containing two capacitors, one inductor, and two independent sources. **CO1, CO3** **6**



- C)** Explain about classification of filters. Draw the characteristics of Low-pass filters. **CO1, CO3** **6**

***** End *****